

At the end of this worksheet you should be able to

- calculate stress and strain and use Young's Modulus to solve for an unknown.
- plot all relevant quantities of simple harmonic motion over time.
- use the quantities of simple harmonic motion and the mathematical description to solve for an unknown.

1. Consider a wire that is 0.1 mm in diameter and 2 m long and a Young's Modulus of 120 GPa ($1 \text{ GPa} = 10^9 \text{ Pa}$). If you applied 100 N to this wire, then what is the stress on the wire? What is the strain? By how much does its length change? What is its new length? What percent change is this?
2. If instead a 200 N force or a 1000 N force is applied then what is stress, strain, length change and percent change in the length from its original length?
3. If instead the wire was 1 m long with 100 N applied, then how much does it stretch? What percent change is this?

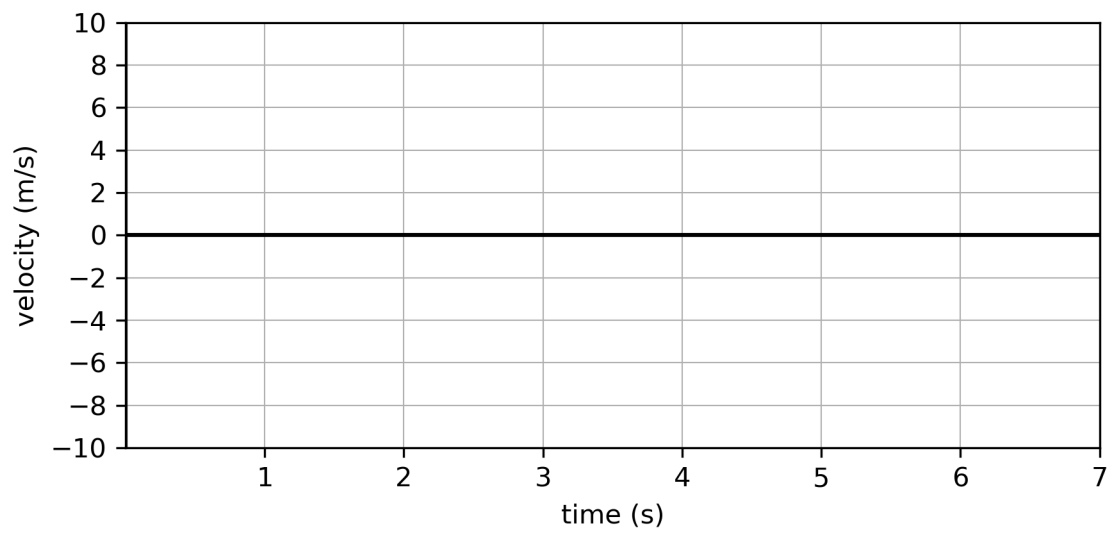
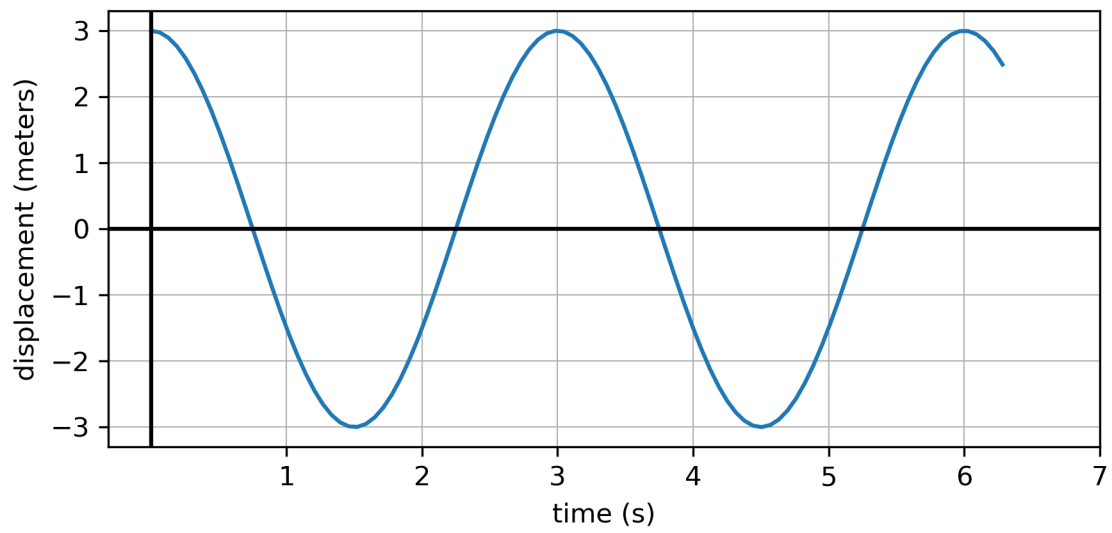
7. If instead of a *parallel* arrangement, we attach one spring to another spring and then to the object, we say that these springs are connected *in series*. Let's find an expressions for an effective spring constant for this arrangement. Each spring will stretch a different amount based on its spring constant, but the object will experience one force and *both of the springs is exerting the same force*.
8. A wire of length l_1 and volume V and cross sectional area A_1 is stretched out to length l_2 , what is its new cross sectional area? Think about this in terms of proportionality.
9. A 60 kg person upright. By how much does the femur shorten if each femur carries half the weight of the person? The cross sectional area of a femur is about 4 cm^2 and the length is about 30 cm Also find the percent change in length.

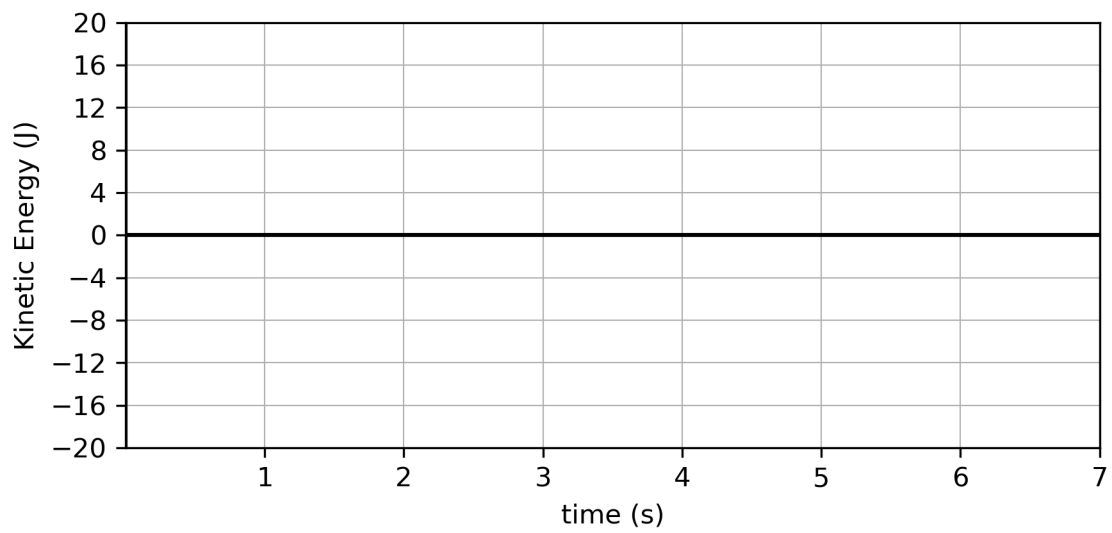
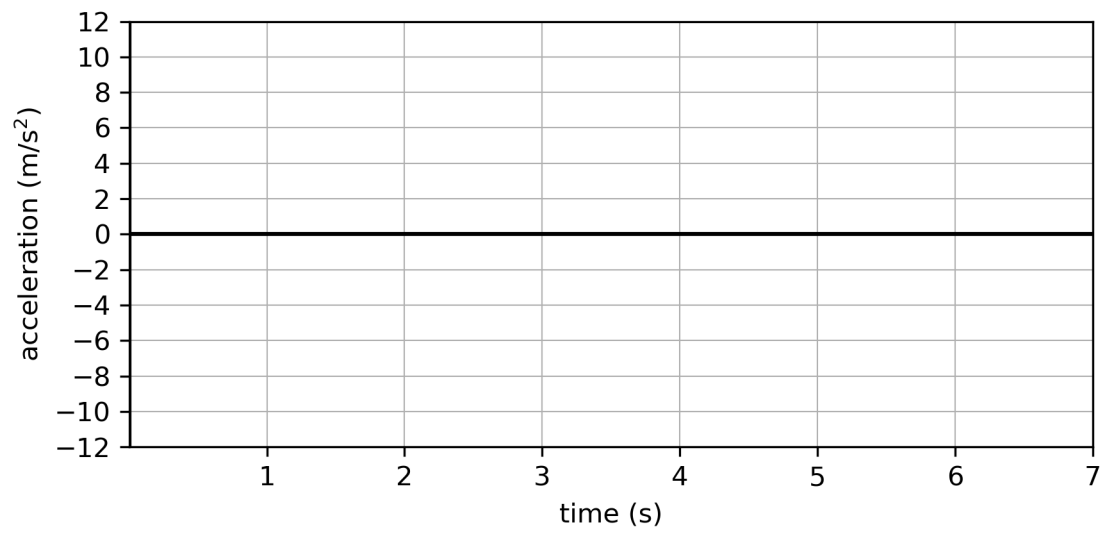
10. A 1 kg mass attached to a spring of spring constant 1000 N/m is positioned so that the spring is stretched 10 cm from its relaxed length. After you finish the following questions, make sure you can write down expressions for all of them in general as well as working them inside out.

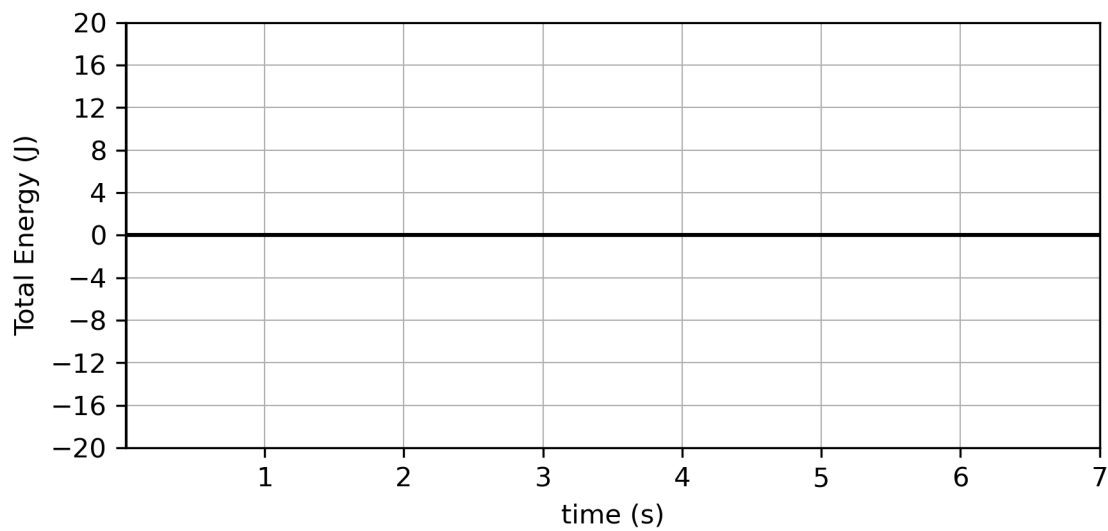
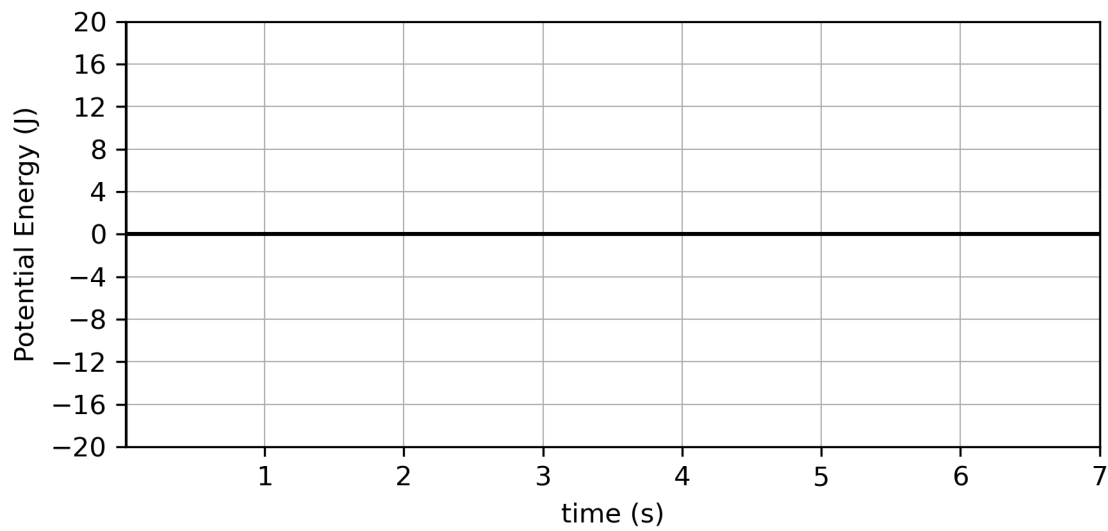
- How much spring potential energy does it have at this position?
- When it is released, and it heads back toward equilibrium gaining kinetic energy as it does, then what is the maximum kinetic energy it can achieve? How stretched out is the spring when it has this much kinetic energy?
- What is the mass's velocity when it has maximum kinetic energy?
- What is the period of this mass's motion? What is its frequency? What is its angular frequency?
- What is the force on the mass when it is at its maximum displacement?

- What is the force on the mass when it is at the equilibrium position?
- What is the maximum acceleration of the mass? Where does this occur?
- Where is the mass when its potential energy and kinetic energy at that position are equal?

11. A 1 kg mass on a spring has the displacement graph that follows. What is the angular frequency, natural frequency, period, spring constant, amplitude, maximum velocity, maximum acceleration, maximum kinetic energy, maximum potential energy, and total energy? Also fill out the rest of the graphs.







12. According to the table below, which material stretches more, 2 m of steel or 1 m of copper of the same width?

13. Four brass wires are subjected to the same tensile stress. The wires have the following unstretched lengths and widths. Rank them in order from least to most change in length.
- (a) length L , diameter d
 - (b) length $2L$, diameter d
 - (c) length $4L$, diameter $d/2$
 - (d) length $L/4$, diameter $d/2$
14. A 0.5 m long guitar string of cross sectional area of $1.0 \times 10^{-6} \text{ m}^2$ and Young's modulus $Y = 2.0 \text{ GPa}$. By how much must you stretch the string to obtain a tension of 20 N.
15. Young's modulus is like the spring constant for materials, but if you stretch a material beyond a limit then the material will not exactly return to its original length. This point of stress is called the *elastic limit* of a material. After this point, *plastic deformation* begins to occur which means the material will be permanently deformed. More stress can be applied beyond this but the material will fracture and break when the stress reaches the *breaking point*. A hair breaks under a tension of 1.2 N and the tensile stress of the breaking point is 200 MPa. What is the diameter of the hair?

16. A copper wire of length 3.0 m is observed to stretch by 2.1 mm when a weight of 120 N is hung from the end. What is the diameter of the wire and what is the stress in the wire? If the breaking point of copper is 400 MPa, what is the maximum weight that may be hung from this wire?

Table 10.1 Approximate Values of Young's Modulus for Various Substances

Substance	Young's Modulus (GPa)
Rubber	0.002–0.008
Human cartilage	0.024
Human vertebra	0.088 (compression); 0.17 (tension)
Collagen, in bone	0.6
Human tendon	0.6
Wood, across the grain	1
Nylon	2–6
Spider silk	4
Human femur	9.4 (compression); 16 (tension)
Wood, along the grain	10–15
Brick	14–20
Concrete	20–30 (compression)
Marble	50–60
Aluminum	70
Cast iron	100–120
Copper	120
Wrought iron	190
Steel	200
Diamond	1200

